

DR. MARIA BEATRIZ WALTER COSTA

EU permanent resident (Swiss-Brazilian) ◇ Germany-based

waltercostamb@gmail.com

<https://waltercostamb.github.io>

<https://linkedin.com/in/waltercostamb>



SUMMARY

Bioinformatician with PhD and 10+ years' experience in analysis of large-scale biological datasets. Proven track record in generating biologically meaningful insights to support data-driven decisions. Published author (*BMC Bioinformatics*, *JMIR Medical Informatics*) with advanced expertise in **Python, R and statistical modeling**.

WORK EXPERIENCE

Friedrich-Schiller University Jena

Research Associate (fixed-term)

Jena, Germany

10/2022 - 09/2025

- Led project on large-scale microbial omics data to identify environmental biomarkers (ML, Python)
- Designed scalable **high-throughput pipeline** for genomics data (extracts **150k+ features from 13k+ genomes**), leveraging parallel computing on high-performance clusters (Snakemake)
- Co-supervised a doctoral researcher and contributed to teaching in large-scale NGS data analysis

University of Leipzig Medical Center

Research Associate (fixed-term)

Leipzig, Germany

02/2020 - 12/2021

- Applied statistical methods to clinical data (**260k+ patients, 20M+ laboratory values, 4 publications**) to support decision in automated diagnosis of 3 acute conditions (Statistics, R)
- Worked closely with physicians and industry partners to build AMPEL Decision Support System
- Structured a clinical knowledge dataflow to improve downstream analysis (SAP-based data)

Brazilian Agricultural Research Corporation (EMBRAPA)

Consultant (fixed-term)

Brasília/DF, Brazil

11/2018 - 03/2019

- Supported strain-engineering decisions by analyzing NGS data (Illumina HiSeq) to find key genes to optimize ethanol production (R, DESeq2)

University of Leipzig

Research Associate (fixed-term)

Leipzig, Germany

04/2017 - 03/2018

- Developed a **new statistical method**, SSS-test (*BMC Bioinformatics*), analyzed 87k structures and identified 14 biomarkers with potential relevance to psychiatric disorders

EDUCATION

Dr. rer. nat. in Computer Science (*magna cum laude*)

University of Leipzig, Chair of Bioinformatics

Leipzig, Germany

2013 - 2018

- Specialization in method development in Computational Biology (6 publications, 3 methods)

M. Sc. in Molecular Biology

University of Brasília, Graduation Program of Molecular Biology

Brasília/DF, Brazil

2010 - 2012

- Specialization in NGS-based analysis in immunogenomics (1 publication)

B. Sc. in Biological Sciences

University of Brasília, Institute of Biological Sciences

Brasília/DF, Brazil

2006 - 2010

- Specialization in Bioinformatics

LANGUAGES

English: near-native speaker (C2, Cambridge CPE)

Portuguese: Native speaker (C2)

German: Advanced (C1)

Spanish: Advanced (B2/C1)

TECHNICAL SKILLS

Programming: Python, R, Perl, Bash, SQL (MySQL)

Bioinformatics: genomics, transcriptomics, AI/ML: machine learning, statistical modeling

Workflow: Snakemake, Git (GitHub)

Data Processing: Linux/Unix, HPC platforms (Slurm, SGE)

SELECTED PRODUCTS

- **FxTractor:** reproducible high-throughput pipeline for feature extraction from microbial genomes (150k+ features across 13k+ genomes), leveraging parallel computing on HPC platforms

<https://github.com/MGXlab/FxTractor>

- **SSS-test:** new method that detects adaptive selection in ncRNAs (*BMC Bioinformatics*) integrating sequence and structural information

<https://github.com/waltercostamb/SSS-test>

SELECTED PUBLICATIONS

- **Walter Costa MB et al.** The Clinical Decision Support System AMPEL for Laboratory Diagnostics: Implementation and Technical Evaluation. *JMIR Med Informatics*, 2021

<https://medinform.jmir.org/2021/6/e20407>

- **Walter Costa, MB et al.** SSS-Test: A novel test for detecting selection on the secondary structures of non-coding RNAs. *BMC Bioinformatics*, 2019

<https://doi.org/10.1186/s12859-019-2711-y>